Ex. No: 2

Title: Covariance, Eigenvalues and Eigenvectors Analysis

Packages: NumPy, Pandas, Matplotlib

AIM:

To find the covariance matrix of a dataset and calculate its eigenvalues and eigenvectors.

ALGORITHM:

1. Start the program and import NumPy, Pandas and Matplotlib.
2. Create a two-dimensional dataset (X and Y) and store it in a Pandas DataFrame.
3. Find the covariance matrix using df.cov().
4. Find the eigenvalues and eigenvectors using np.linalg.eigh().
5. Plot the data points and the eigenvectors.
6. Display the output and stop.

PROGRAM:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(0)
x = np.random.randn(100)
y = 2 * x + np.random.randn(100) * 0.5
df = pd.DataFrame({"X": x, "Y": y})

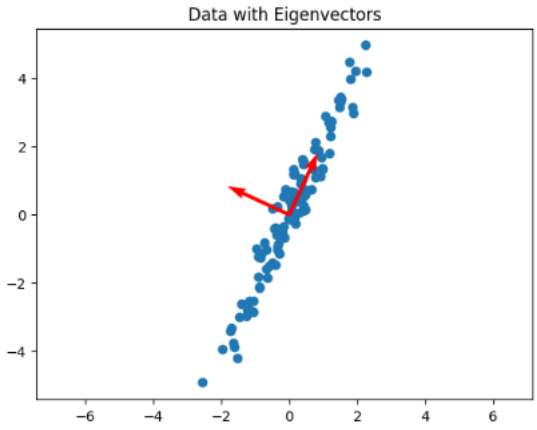
cov = df.cov()
print("Covariance matrix:\n", cov)

values, vectors = np.linalg.eigh(cov)
print("Eigenvalues:", values)
print("Eigenvectors:\n", vectors)

plt.scatter(df["X"], df["Y"])
for i in range(2):
    plt.quiver(0, 0, vectors[0, i] * 2, vectors[1, i] * 2,
               angles='xy', scale_units='xy', scale=1, color='red')
plt.axis("equal")
plt.title("Data with Eigenvectors")
plt.show()
```

OUTPUT:

```
Covariance matrix:
          X         Y
X  1.026087  2.111020
Y  2.111020  4.610069
Eigenvalues: [0.04902823 5.58712782]
Eigenvectors:
 [[-0.90751033  0.42002977]
 [ 0.42002977  0.90751033]]
```



RESULT:

The covariance matrix, eigenvalues and eigenvectors were calculated successfully. The eigenvector with the largest eigenvalue shows the direction of maximum variance.